Intermediate Physics Lab 2: AC current rectification

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Abstract

An abstract is a short paragraph summary (¡ 150 words). Abstract overviews what the report entails from what you did, what you found (measured), and what you conclude. In general, start abstracts with 'We report on...' followed by why you did it (here you can state why your specific experiment is justified for the particular lab module), what you found, and what you conclude. The abstract should grab the attention of a casual reader so that they want to read the full manuscript in detail. Keep it short and simple.

1 Introduction

The goal of this experiment is to verify if the rectification circuit performs well, which converts an AC signal into a DC signal. A basic form of such circuit is as follows.

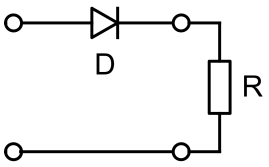


Figure 1: Basic half-wave rectifier

A half-wave rectifier is less efficient than a full-wave rectifier, as it only uses one half of the AC cycle, leading to lower output voltage and higher ripple, requiring more filtering.

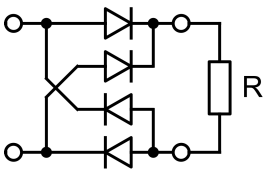


Figure 2: Basic full-wave rectifier

In contrast, a full-wave rectifier makes use of both halves of the AC cycle, providing higher output efficiency and a smoother DC signal with less ripple.

It is reasonable to assume the resistance of diode, denoted by r. In this experiment, various elements were used to check if it was converted into DC well. In particular, resistance, capacitor or diodes were used to verify characteristics of the signal. If there are no elements, but just linked with cable, the output voltage is as follows.

$$V_{NULL} = V_0 \left|\sin(\omega t)\right| \tag{1}$$

If the element is a resistance R,

$$V_R = \frac{R}{2r+R} V_0 \left|\sin(\omega t)\right| \tag{2}$$

If the element is a capacitor C,

$$\frac{d}{dt}V_C + \frac{1}{RC}V_C = \frac{V_0}{RC}\left|\sin(\omega t)\right| \tag{3}$$

While the absolute of sin function is as follows.

$$|\sin(x)| = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos(2kx)}{4k^2-1}$$

Therefore,

$$V_C = \frac{2}{\pi}V_0 - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{V_0}{4k^2-1}\mathrm{Re}\left(\frac{e^{2ik\omega t}}{2ik\omega RC+1}\right) \tag{4}$$

In this experiment, the voltage drop across diodes plays a crucial role in the performance of the rectification circuit. When constructing a full-wave rectifier using four diodes, each diode introduces a forward voltage drop, typically around 0.7V for silicon diodes,

which affects the overall output voltage of the rectified signal. In the case of using LEDs, the forward voltage drop is even more significant, ranging between 1.8V and 3.3V, depending on the LED's material and color. The objective of this experiment is to evaluate how these voltage drops influence the effectiveness of the rectification process, particularly in terms of the amplitude and smoothness of the rectified DC output. Understanding the voltage drop behavior is essential for optimizing the rectifier circuit, as it directly impacts the power efficiency and the quality of the DC signal delivered to the load.

2 Experiment

2.1 Apparatus

The rectification circuit is constructed by four diodes, working in AC voltage provided by Analog Discovery 2 ([2]). The frequency is varied from $0.02Hz$ to $200Hz$ in logarithmic scale, at 40 sampling points. For each frequency experiment, the output voltage of exactly ten period of each drive voltage is measured. We wanted to modify the capacitance of the circuit, used 4 $4.7\mu F$ and 1 $100\mu F$ capacity capacitor. The serially connected capacitor is examined, five different situation is tested. True capacitance value of each component is measured by multimeter, and its result is presented at Fig. 3. By the true value measured, the circuit capacity of $90.86\mu F$, $4.622\mu F$, $2.255\mu F$, $1.478\mu F$, and $1.115\mu F$ be the circuit capacity condition of each experiments.

Index	Measured Capacitance
4.7 #1	$4.622 \pm 0.002\mu F$
4.7 #2	$4.405 \pm 0.002\mu F$
4.7 #3	$4.288 \pm 0.002\mu F$
4.7 #4	$4.543 \pm 0.002\mu F$
100 #1	$90.86 \pm 0.02\mu F$

Figure 3: The measured capacitance of each capacitor component. The index is labeled by the factory value denoted in arbitrary order.

2.2 Methods

Construct the circuit as (figure), and plug drive voltage to the input node respectively. Connect the nodes between drive voltage in Channel 1, and output voltage in Channel 2 of Analog Discovery 2 Oscilloscope. Turn on the sinosodic signal of $2.5V$ and modified frequency in order to the drive voltage, and wait for 1 second for the circuit to be saturated. Measure 100 points per period for sampling, for 10 period duration. Grab the initial time data and construct time grid for each measured result.

The machine works by the python code built, with SDK provided by [2]. The code for such frequency sweep experiment is denoted in 5.1.

3 Results and Conclusions

Matlab's calculation results are also taken into the consideration to compare with acquired data. The code used for the calculation is attached on the last page. (will be added)

4 Conclusions

In this section, discuss the significance of your results in relation to what you proposed in the introduction.

5 Appendix

5.1 Codes for AD2

The code and relavent wrapper library is open to public in [1].

```
from pydwf import DwfLibrary
from pydwf.utilities import openDwfDevice
import numpy as np
from libs.pydwf_wrapper import
    generate_sine_wave, measure_voltage
import matplotlib.pyplot as plt
import time
from alive_progress import alive_bar

dwf = DwfLibrary()
def experiment(experiment_config):
    create_folder(results_folder)
    with openDwfDevice(dwf) as device:
        total_results = []
        amplitude = 5
        freq_ini = 0.02
        freq_fin = 0.2
        num = 10

        with alive_bar(num) as bar:
            for frequency in np.linspace(
                freq_ini, freq_fin, num):

                output_channel = device.
                    analogOut
                generate_sine_wave(
                    output_channel, frequency=
                        frequency, amplitude=
                        amplitude
                )

                time.sleep(1)
                input_channel = device.
                    analogIn
                result = measure_voltage(
```

```
                    input_channel,
                    record_time=1 / frequency
                        * 10,
                    sampling_frequency=
                        frequency * 100,
                    number_of_channel=2,
                )

                config_dict =
                    experiment_config.copy()
                config_dict["frequency"] =
                    frequency
                config_dict["amplitude"] =
                    amplitude
                result["config"] = config_dict

                total_results.append(result)
                bar()

    return total_results
```

Listing 1: Python Code Example

6 References

References

[1] Woojin Han. intermediate physics experiment 2. https://github.com/WoojinHan24/intermediate_phys_experiment_2, 2024. [Online].

[2] S. K. Analog discovery 2 (legacy). https://digilent.com/reference/test-and-measurement/analog-discovery-2/start, n.d. Digilent Reference.